The temporal dynamics of infectious diseases

Background

Infectious diseases are ubiquitous in natural populations and pose great challenges to hosts. Infectious diseases are know to shape not only the well-being of hosts, but could also cause mortality and eventually population declines and extinctions. For instance, Chytridiomycosis is an infectious diseases in amphibians that is caused by the fungus *B. dendrobatidis*. Chytridiomycosis has been reported to cause dramatic population declines and species extinctions in amphibians all over the world and is assumed to further expand into novel areas and to infect new species. In this study project, you will study the temporal dynamics of infectious diseases by the use of individual-based simulations. These simulations should allow you to get a better mechanistic understanding of such dynamics and inspire further explorations.

The population

We study a single host species with population size $N=10{,}000$ that gets infected by a single disease (e.g., a virus, fungi, bacteria, etc). The disease is not causing any mortalities and the disease dynamics are much faster than the demographic changes in the host. Therefore, you should assume that N is constant throughout the simulations. Each host individual can be in one of three distinct states:

- *susceptible*
- *infected*
- *recovered*

Susceptible individuals do not posses any resistance against the disease and can be infected during contact with an infected individual. *Infected* individuals carry the disease and can infect susceptible individuals. *Recovered* individuals survived the disease, do not carry the disease anymore, and developed full resistance. As a consequence, recovered individuals cannot get infected anymore and cannot transfer the disease to susceptible individuals. Here, we assume that resistance against the disease stays forever.

We assume that all host individuals are located in a single population, where each individual has the same probability to meet any other individual. In other words, the host population does not show any social or spatial structure. Throughout these simulations, we further assume that the disease enters the host population at very low densities when only a few individuals are infected at the beginning of the simulation. This can happen when the disease enters the population through dispersal, or through *de novo* mutations of already existing, less infectious variants.

Note that these simulations will be very close to so called SIR models (**s**usceptible-**i**nfected-**r**esistant; which are usually studied in continuous time using differential equations). In contrast to "classic" SIR models, however, you will account for demographic stochasticity and model disease dynamics in discrete time steps (e.g., days).

The daily events

Each day, the following events occur consecutively:

1. recovery

In the early morning of each day, during the sleep of the host, host individuals recover from the disease with probability γ and then enter the "recovered" state.

2. novel infection

During daytime, when the host individuals are awake and hunt for food or look for mating partners, each infected host individual gets into contact with a fixed number of other host individuals (c, the number of "potentially infectious contacts" per day). These "contacts" are picked randomly from the whole population for each day anew without replacement. Infected individuals transmit the disease to susceptible individuals with a certain probability (α, the probability that a single contact between an infected individual and a susceptible individual leads to infection of the susceptible).

3. storage of summary statistics

At the end of each day, store meaningful summary statistics. This includes the number of hosts in the susceptible, infected, and recovered state.

Study the disease dynamics systematically

After finishing debugging, study the disease dynamics systematically. To this end, choose a proper study design and collect meaningful summary statistics. At the end of this systematic evaluation (i.e. exploration of the parameter space), you should be able to say how the transmissability (α), the recovery rate (γ), and the number of contacts (c) affect the course of the disease dynamics. In particular, how do these characteristics determine

a) the proportion of the population that got infected over the course of the pandemic

b) the maximum number of infected individuals at one point in time (the "height of the wave")

You can quantify even more aspects of the disease dynamics (other than *a* and *b*) if they are meaningful. Importantly, a mathematical analysis of this model predicts that a disease could only spread and infect a large number of people when the base reproductive number R_0 is greater than 1, with

$$R_0 = \frac{c\ \alpha\ (1-\gamma)}{\gamma}.$$

What does R_0 mean? And can you confirm this expectation?